INCH-POUND

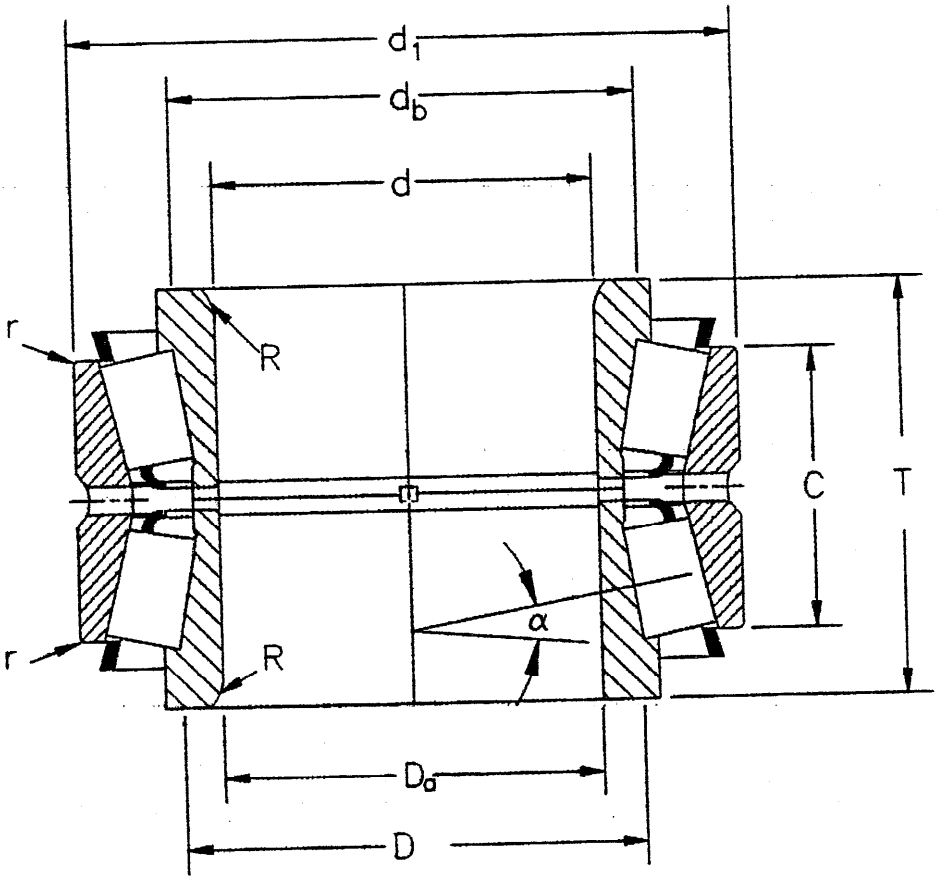
FF-B-187/11
10 SEPTEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, ROLLER, TAPERED, DOUBLE ROW, STRAIGHT BORE, TWO SINGLE CONES, CONE BACK FACE RIB GROUND ON THE OUTER DIAMETER FOR SEALING PURPOSES, ONE DOUBLE CUP LUBRICATION HOLES AND GROOVE. ('TYPE TNASWE)

This specification is approved by the Adminstrator, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

The complete requirements for procuring the bearings described herein, shall consist of this document and the latest issue of FF-B-187/GEN.



BEARING CONFIGURATION

FSC 3110

AMSC N/A

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	PART NUMBER 1/		d	D	T	C	R 2/	r 2/	db	Da	K FACTOR	BASIC DYNAMIC LOAD RATING (LBS)
	CONE	CUP	BORE	OUTSIDE DIA	OVERALL WIDTH	OVERALL CUP WIDTH	MAX SHAFT FILLET RAD	MAX HOUSING FILLET RAD	RECOMMENDED SHOULDER DIA			
									SHAFT	HOUSING		
01	NA05076-SW	05185D	0.7500	1.8504	1.3750	0.9926	0.03	0.03	0.94	1.67	1.64	7500
02	NA15117-SW	15251D	1.1813	2.5000	2.0001	1.4375	0.03	0.03	1.40	2.32	1.67	14000
03	NA24776-SW	24720D	1.5000	3.5000	2.3125	1.5625	0.03	0.03	1.77	2.83	1.49	2100
04	NA439-SW	432D	1.7500	3.7500	2.6250	2.0000	0.03	0.03	2.05	3.43	2.05	35200
05	NA435-SW	432D	1.7500	3.7500	2.8126	2.0000	0.14	0.03	2.24	3.43	2.05	35200
06	NA456-SW	452D	2.0000	4.2500	2.9376	2.1250	0.14	0.03	2.56	3.94	1.74	37400
07	NA483-SW	472D	2.7559	4.7244	2.9376	2.1250	0.14	0.03	3.27	4.49	1.52	39600
08	NA580-SW	572D	3.2500	5.5115	3.6250	2.6250	0.14	0.03	3.86	5.24	1.45	51100
09	NA497-SW	493D	3.3750	5.3750	2.8750	2.1250	0.14	0.03	3.90	5.12	1.31	42800
10	NA596-SW	592D	3.5000	6.0000	3.6250	2.5000	0.14	0.03	4.09	5.67	1.32	59700
11	NA56425-SW	56650D	4.2500	6.5000	3.5000	2.5000	0.14	0.03	4.84	6.26	1.18	62700
12	NA48290-SW	48220D	5.0000	7.1875	3.6874	2.8750	0.14	0.03	5.55	6.93	1.91	74100
13	NA48685-SW	48620D	5.6250	7.8750	3.6876	2.8750	0.14	0.03	6.22	7.60	1.74	74600
14	NA46790-SW	46720D	6.5000	8.8750	3.7500	2.7500	0.14	0.03	7.13	8.58	1.52	78200
15	NA87700-SW	87112D	7.0000	11.1250	4.2500	3.1250	0.14	0.06	7.87	10.50	1.41	119700
16	LM637349-NW	LM637310D	7.2500	9.5625	3.7500	2.7500	0.14	0.06	7.83	9.29	1.39	86500
17	NA67885-SW	67820CD	7.5000	10.5000	4.3125	3.3125	0.14	0.06	8.23	10.20	1.22	115900
18	LM241149-NW	LM241110D	8.0000	10.8750	3.7500	2.8750	0.14	0.06	8.66	10.51	1.83	112100
19	LM446349-NW	LM446310D	9.2500	12.2500	4.0000	2.8750	0.14	0.06	9.92	11.85	1.61	130200
20	NA8575-SW	8520CD	9.2500	12.8750	4.6250	3.2500	0.25	0.06	10.20	12.32	1.44	149200
21	LM249747-NW	LM249710D	9.9990	13.6875	4.0000	2.7500	0.14	0.06	10.71	13.11	1.20	105400
22	LM251649-NW	LM251610D	10.5000	13.8750	4.2500	3.2500	0.25	0.06	11.46	13.39	1.83	154800
23	L3577049-MW	L357010CD	12.0000	15.5000	4.2500	3.2500	0.25	0.06	12.95	14.96	1.63	172900

1/ PART NUMBERS ARE FOR REFERENCE ONLY. PART NUMBERS REFLECT THE CUP AND CONE NUMBERS USED BY INDUSTRY AND THE AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA).

2/ THESE MAXIMUM FILLET RADII SHALL BE CLEARED BY THE BEARING CORNERS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Material: Cones, cups, and rollers: bearing quality - carburizing grade alloy or through-hardening grade alloy steel in accordance with ASTM-A295. Cage: carbon steel (stamped). Powdered process steel may be used.

2. Hardness: Cones, cups and rollers: 58 to 64HRC.

3. Tolerances: Standard grade (AFBMA Class 4) (allowable tolerances are in 0.0001 inch).

Cone bore (d)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0 3.0000	3.0000 6.0000	5 10	0 0

Cup diameter (D)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000 12.0000	12.0000 24.0000	10 20	0 0

Bearing width (C)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0 5.0000	5.0000 12.0000	100 300	0 0

Assembled bearing maximum radial runout		
Cup O.D. (D)		Tolerance
Over	Incl.	
0	24.0	20

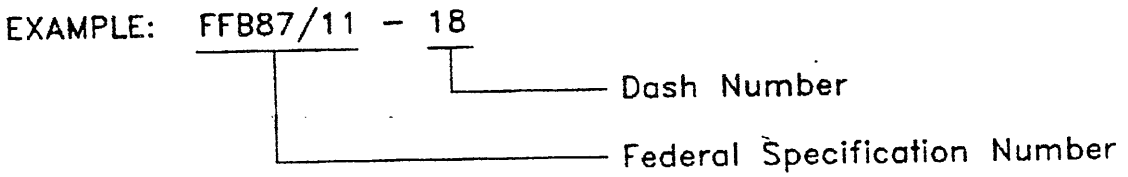
4. Operating temperature: Recommended operating temperature not to exceed 230° F.

5. Lubrication: Bearings shall be furnished without lubrication. All bearings shall be preserved in accordance with MIL-P-197.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:46Z
Check the source to verify that this is the current version before use.

7. Contact angle: All bearings are normal angle bearings having a contact angle (a) between 10 and 19 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

8. Part number: The part number consists of the letter "F", the basic number of this specification sheet, followed by a dash number from Table I.



9. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

10. Referenced documents shall be of the issue in effect on the date of invitation for bids, or request for proposal.

NOTE: inch units can be converted to soft metric units as follows:
inch [X] 25.4 [=] millimeters.

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodian:
Air Force – 11

Review Activities:
Army - AR, MI
Navy - OS
Air Force – 99

User Activities:
Army - GL, ME
Navy - MC, YD

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA - FSS

PREPARING ACTIVITY:
DLA - IS

(Project 3110-0847-11)

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:46Z
Check the source to verify that this is the current version before use.